

Palm-Astro History (Indian Data) Computer Vision + Multimodal Inference

Goal

Build a small AI system that, from a photo of a person's palm (Indian population), **segments major palm lines, extracts geometric features**, and predicts **interpretable palm attributes** such as line dominance, palm type, or simple life event indicators (synthetic labels).

The model should be **data-driven, interpretable**, and demonstrate **feature-based reasoning** — not random outputs.

Target: high accuracy and reproducibility on **~200 annotated Indian palm images**.

Task

1. Use or create a small dataset (~200 Indian palm images).
 - Include **left and right hands** if available.
 - Add **segmentation masks** for Life, Head, and Heart lines.
 - Maintain metadata (hand type, gender, region, age group if known).
2. **Train a segmentation model** (e.g., U-Net / SegFormer) to detect the three major lines.
 - Evaluate using **IoU or Dice score** per line.
3. **Extract geometric features** from the segmentation output, such as:
 - Line lengths, intersections, curvature, aspect ratio, dominant line type.
4. **Build a small predictive or rule-based module** to classify each palm into 1–2 labels, e.g.:
 - “Heart-line dominant” / “Head-line dominant” / “Balanced”

- “Early career shift: yes/no” (synthetic label OK)
- 5. **Create a simple Gradio or Streamlit demo** that:
 - Accepts palm image upload
 - Displays segmentation overlay
 - Shows extracted features and predictions with confidence
- 6. **Ensure interpretability** — results must come from measured features, not hardcoded text.

Suggestions

Dataset

- Use smartphone photos or a small Roboflow project (~200 palms).
- Crop palms clearly (no faces) and label manually (Life/Head/Heart lines).
- Perform small augmentations (rotation, color jitter, blur, scaling).

Modeling

- **Segmentation:** U-Net (EfficientNet encoder) or SegFormer.
- **Features:** Compute line length, curvature, intersection count, palm ratio.
- **Classification:** XGBoost / RandomForest / small CNN.

Evaluation

Metric	Target
Mean IoU (segmentation)	≥ 0.70
Classification F1	≥ 0.65

Explainability

SHAP / feature importance /
Grad-CAM

Inference demo

Must run smoothly

Timeline (2–3 Days)

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Focus

- 1 Data setup & preprocessing
- 2 Model training + feature extraction
- 3 Evaluation + demo + README

Deliverables (GitHub Only)

GitHub Repository (public link) must include:

-  palm-astro/
- |— data/ (or Roboflow link)
- |— train_segmentation.ipynb / train.py
- |— utils/feature_extraction.py
- |— app.py # Gradio or Streamlit demo
- |— results/ (sample overlays + metrics.csv)
- |— requirements.txt
- |— README.md (how to run + dataset + ethics)
- |— evaluation.md (accuracy, analysis, suggestions)

README must cover:

- Dataset source and preprocessing summary
- Model architecture and training approach
- Evaluation metrics (IoU, F1)
- Ethical/data privacy notes
- Commands to run the app
- Example output screenshots